



Griffith College Dublin
ASSIGNMENT TITLE SHEET
FACTULY OF COMPUTING SCIENCE

Course	BSCH
Stage / Year	4
Module	Machine Learning
Semester	2
Assignment	Assignment 1
Date of Title Issue	10.03.2026
Assignment Deadline (Graded)	29.03.2026 @ 23:59
Assignment Submission	Upload to Moodle
Assignment Weighting	30% of the module

Assignment Title

Error Based Learning (Regression)

Artificial Intelligence Assessment Scale (AI AS):

This assignment is subject to AI permission option:

No AI Use ☒ [Y]
AI for Planning ☐ []
AI for Editing ☐ []
AI for Support Tasks ☐ []
AI for Collaboration ☐ []
Full Use of AI ☐ []

If no checkbox is selected, "No AI Use" will be applied. AI Assessment Scale Framework:
<https://www.griffith.ie/sites/default/files/2025-05/f-6.5-ai-in-learner-assessment-policy.pdf>

Please note:

- All AI generated content and process must be declared.
- Any task that is not listed as permitted may be assumed to be prohibited.
- If it is determined that a learner has used an AI tool in any way that is not permitted by the chosen level, then they will be subject to the disciplinary procedures listed in the Academic Integrity and Misconduct policy regarding the misuse of AI in assignments.

Assignment Specific AI Instructions:

All coding should be your original coding, no GenAI coding should be used.

Learning Outcomes

This assignment is assessing the following programme and related module learning outcomes:

1. Implement various data machine learning algorithms and techniques.
2. Analyse machine learning results and determine where improvements can be made.

Assessment Criteria

Assessment criteria applied to this assignment:

1. Clarity of implementation
2. Correctness
3. Discussion and critical analysis

Submission Instructions and Plagiarism

Plagiarism:

All work must be your own. Any cases of suspected plagiarism will be thoroughly investigated and may be brought in for VIVA. All parties involved in cases of plagiarism will receive a 0 and their student record marked accordingly. Any submission that is found to be plagiarised will receive a grade of 0 without the possibility of resubmission.

By submitting your assignment you accept that you understand what plagiarism is and that your assignment is not plagiarised in any way.

Instructions for submission:

All work must be submitted via the submission point on Moodle, email submissions not accepted.

In line with good academic practice, you must carefully acknowledge all sources including any material sourced from other students. You must fully reference the source(s) of such work so that examiners may assess the value of your contribution. You may be required to explain/present your submission as part of the work. For that reason it is important that you understand what you present.

The use of AI should be clearly described at the end of your submission

Penalties

Late Penalties:

Standard penalties will be applied to work that is submitted late, as per faculty guidelines.

You might randomly be selected for live presentation (VIVA)

Machine Learning

Assignment 1¹

30% of the module

Error Based Learning (Regression)

Due on Sunday 29/03 at 23:59

You will use the Insurance Charges Dataset (from Kaggle)². This dataset includes information about insurance charges and various features that might influence these charges, such as age, sex, BMI, number of children, smoker status, and region.

1. Preprocess the data
2. Implement the Linear Regression:
 - a. Implement the Linear Regression algorithm using Python without using any machine learning libraries (e.g., scikit-learn). You can use libraries for mathematical operations (e.g., NumPy).
 - b. Implement the gradient descent algorithm to find the optimal weights.
 - c. Include a function to compute the Mean Squared Error (MSE).
3. Train and evaluate your model
4. Compare with Linear Regression implemented in scikit-learn library
5. Analysis and Reporting

Submission:

A Jupyter Notebook that includes all your codes, visualizations and analysis. It should also include your discussion and comment.

Notes:

1. Standard late penalty applies.
2. Make sure it is your own original work.
3. Submission is through moodle.

¹ Edited by Abubakr Siddig on 05.03.2026

² <https://www.kaggle.com/datasets/thedevastator/prediction-of-insurance-charges-using-age-gender>